



UNIVERSITÄT
LEIPZIG

Asymmetric Numeral System

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Outline

1. Introduction
2. Asymmetric Numeral System
3. rANS
4. Streaming-rANS
5. Summary
6. Reference

Introduction

Outline

1. Introduction

- 1.1 Background
- 1.2 Information Theory
- 1.3 Source Coding Theorem
- 1.4 Entropy Coding

- 1948: Claude Shannon introduced the concept of information theory in his paper "A Mathematical Theory of Communication" [1].
- 1950: David Huffman introduced Huffman coding [3].
- 1977: Arithmetic coding was introduced by Robert M. Fano [4].
- 2014: Asymmetric Numeral System (ANS) was introduced by Jarosław Duda [2].
- 2016: Zstandard (Zstd) was introduced by Yann Collet [5].
- 2018: AV1 Image File Format (AVIF) was introduced by Alliance for Open Media [6].
- 2020: JPEG XL was introduced by JPEG Committee [7].

Information Theory^[1]

- Claude Shannon: "A Mathematical Theory of Communication"
- Self-information
a symbol of probability p contains $\log_2 \frac{1}{p}$ bits of information.
- Shannon Entropy
Let X be a random variable, which takes values in the set $\mathcal{X}$ and is distributed according to p

$$H(X) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{1}{p(x)}$$

Source Coding Theorem (Shannon) ^[1]

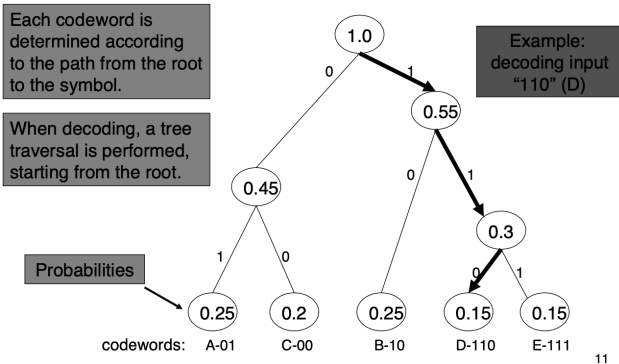
- prerequisite: lossless data compression
- $H(X)$ is compression limit, i.e. lower bound $L_{\text{avg}} \geq H(X)$
which measures the minimum average number of bits needed to encode a symbol from a data source
- $H(X)$ is achievable, $L_{\text{avg}} \rightarrow H(X)$
for a sufficiently long sequence of symbols it is possible to achieve an average code length close to $H(X)$
- Shannon's Source Coding Theorem ^[1]

$$N \cdot H(X)$$

Entropy Coding

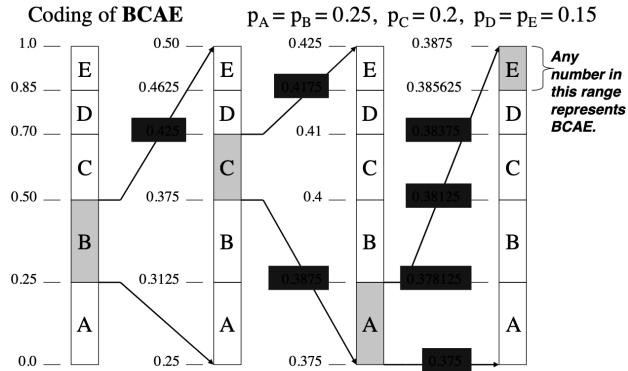
- Implementation of Shannon's Source Coding Theorem
- High-probability symbols get shorter codes
- Low-probability symbols get longer codes.
- A method that attempts to approach the lower bound $H(X)$
- Huffman Coding: ZIP, PNG
- Arithmetic Coding: JPEG, H.264, HEVC.
- Asymmetric Numeral System (ANS): Zstandard, AVIF, JPEG XL

Huffman Coding



source: <http://www.math.tau.ac.il/~dcor/Graphics/adv-slides/entropy.pdf>

Arithmetic Coding



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source: <http://www.math.tau.ac.il/~dcor/Graphics/adv-slides/entropy.pdf>

Asymmetric Numeral System

Outline

2. Asymmetric Numeral System

2.1 Motivation

2.2 Key Concepts

2.3 Performance

Motivation

- Huffman coding is efficient for symbol-based compression but cannot achieve optimal compression in many cases due to its restriction to integer-length codes
- Arithmetic coding achieves near-optimal compression but has a higher computational cost, especially in encoding and decoding (it uses multiplication and division operations for each encoded symbol)
- ANS combines the compression ratio of arithmetic coding (integer arithmetic, modular arithmetic, and table lookups), with a processing cost similar to that of Huffman coding ^[4]

Key Concepts

- **Symbol spread function** → "Numeral System", e.g. standard binary system

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- **Representation of information**, e.g. standard decimal system
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- **Symbol spread function** → "Numeral System", e.g. standard binary system
- **Representation of information**, e.g. standard decimal system
represents an entire message as a single number
- **Symmetric or "Asymmetric"**

Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols ^[3]

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- $\bar{s} : \mathbb{N} \rightarrow \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$

Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols ^[3]
- $\bar{s} : \mathbb{N} \rightarrow \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$
- In the standard binary system example: $\bar{s}(x) = \text{mod}(x, 2)$
i.e. splits natural numbers into subsets of even/odd numbers ^[3]

Symbol spread function (Exa. 1)

$$S = \{0, 1\}$$

Exa. 1: Standard Binary System

$\mathbb{N}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$s = 0$																			
$s = 1$																			

Fig.1

Symbol spread function (Exa. 2)

$$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$x' = 10x + s$$

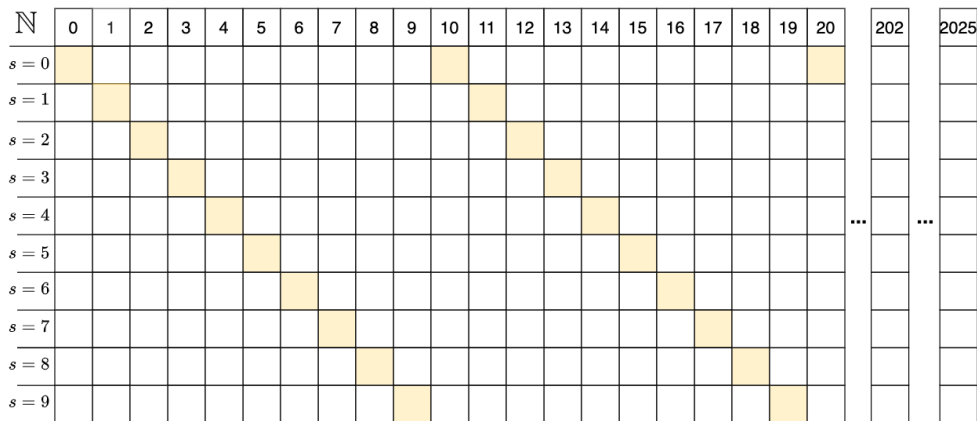


Fig.2

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

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$$S_{in} = (2, 0, 2, 5)$$

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- the sequence of symbols/digits to be encoded:

$$S_{in} = (2, 0, 2, 5)$$

- How can we represent this sequence as a single number?

The easiest way is to concatenate the symbols/digits:

$$x_{out} = 2025$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$x_{\text{init}} = 0$$

$$x_1 = x_{\text{init}} \times 10 + 2 = 2$$

$$x_2 = x_1 \times 10 + 0 = 20$$

$$x_3 = x_2 \times 10 + 2 = 202$$

$$x_{\text{out}} = x_3 \times 10 + 5 = 2025$$

Representation of Information (Exa. 2)

- What is the encoding function?

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- Encoding function

$$x_{i+1} = x_i \times 10 + s_i$$

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$$x_{i+1} = x_i \times 10 + s_i$$

- x_{i+1} is the x_i -th appearance of symbol s ^[3]

$$x_{i+1} = C(s, x_i) = x_i \cdot 10 + s$$

Representation of Information (Exa. 2)

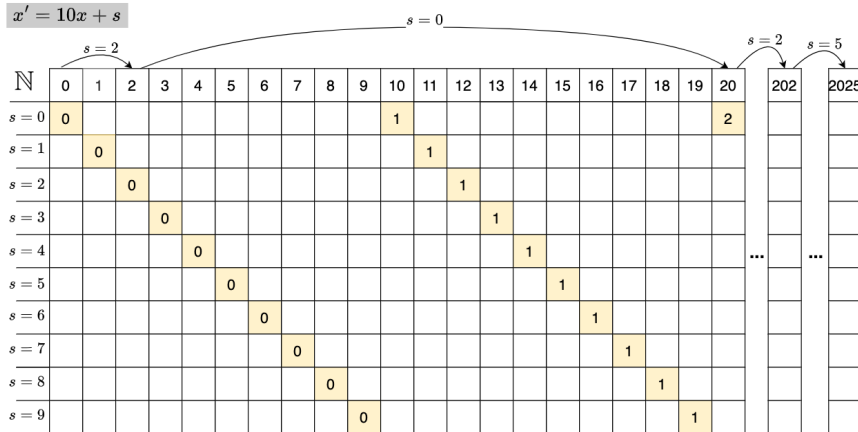


Fig.3

Representation of Information (Exa. 1)

Exa. 1: Standard Binary System

$$x_{i+1} = 2x_i + s$$

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x_{i+1}	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$s = 0$	0		1		2		3		4		5		6		7		8		9
$s = 1$		0		1		2		3		4		5		6		7		8	

Fig.4

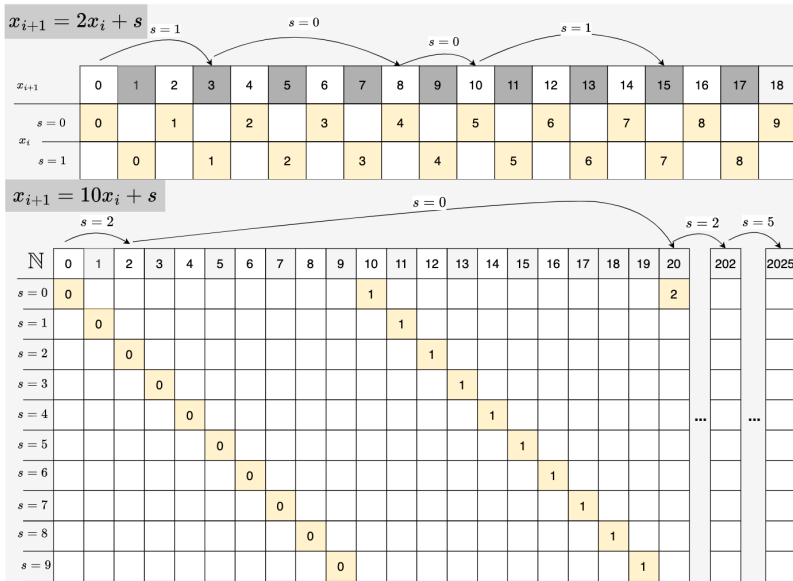


Fig.5

What is asymmetric?

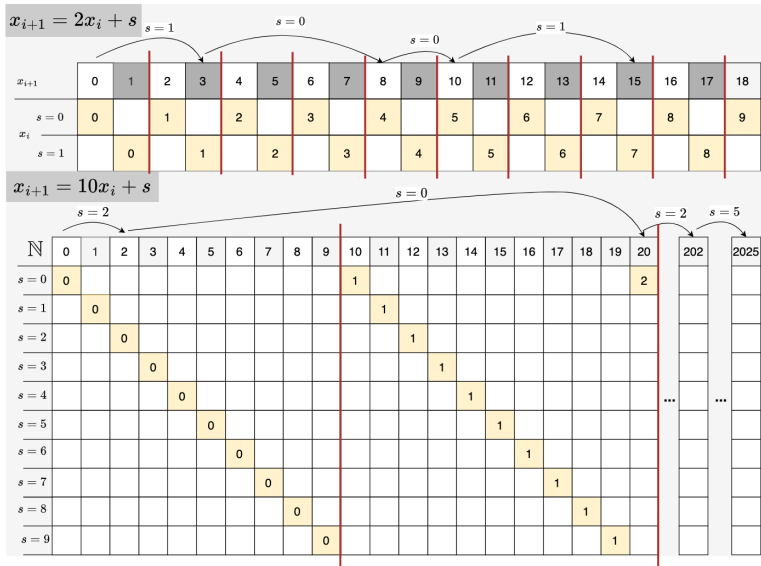


Fig.6

Exa. 3: Asymmetric Binary System ^[3]

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$

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Exa. 3: Asymmetric Binary System ^[3]

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$
- a cycle of length 4, which contains 1 zero and 3 ones
- Assumption:

$$x_{i+1} = C(s, x_i) \approx x_i/p_s, \text{ where } p_s \text{ is the probability of symbol } s$$

Exa. 3: Asymmetric Binary System ^[3]

$$x_{i+1} \approx x_i / Pr(s) \quad Pr(0) = 1/4, Pr(1) = 3/4$$

x_{i+1}	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	...	52
$s = 0$																					
x_i																				...	
$s = 1$																					

Fig.7_0

Exa. 3: Asymmetric Binary System ^[3]

$$x_{i+1} \approx x_i / Pr(s) \quad Pr(0) = 1/4, Pr(1) = 3/4$$

x_{i+1}	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	...	52
$s = 0$																				...	
x_i																				...	
$s = 1$																				...	

Fig.7_1

Exa. 3: Asymmetric Binary System ^[3]

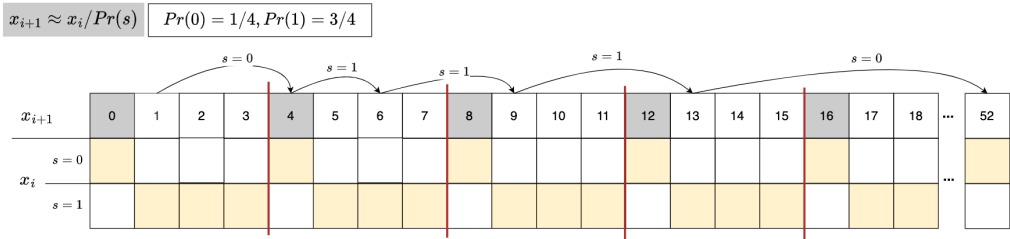


Fig.7_2

Exa. 3: Asymmetric Binary System [3]

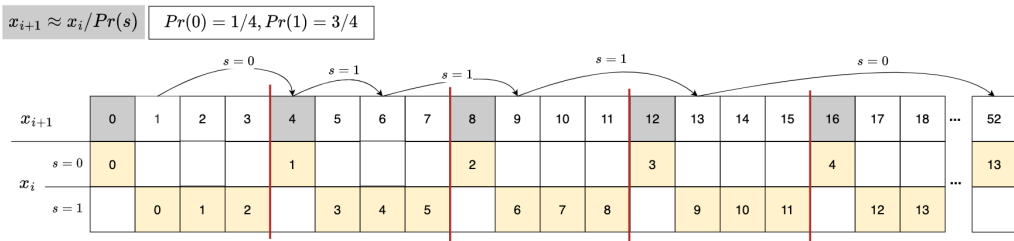


Fig.7

We refine even/odd number by changing their densities

Asymmetric Numeral System | Key Concepts | Symmetric or Asymmetric

Symmetric or Asymmetric

- Symmetric: uniform distribution
- Asymmetric: non-uniform distribution

Symmetric or Asymmetric

Assumption: $x_{i+1} = C(s, x_i) \approx x_i/p_s$, where p_s is the probability of symbol s

- To change the symmetric behavior so that the information added to x_i by coding a new symbol depends on the probability distribution of the symbols

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- A symbol of probability p carries $\log_2(1/p)$ bits of information (Shannon)

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- A symbol of probability p carries $\log_2(1/p)$ bits of information (Shannon)
- How many bits of new information? $\log_2 x_i + \log_2(1/p) \rightarrow \log_2 x_{i+1}$
- $\log_2 x_{i+1} = \log_2 x_i + \log_2(1/p) = \log_2(x_i/p) \Rightarrow x_{i+1} \approx x_i/p$

Performance

— f

rANS

Outline

3. rANS

3.1 Key Concepts

3.2 101110

3.3 Notations

3.4 Encoding Function

3.5 Implementation of Encoder

3.6 Decoding Function

3.7 Implementation of Decoder

rANS

- "r" = "range": similar to range coding (RC)
- symbol spread function
- representation of information
- asymmetric

Exa.1: Standard Binary System

- How to encode 101110? What is the encoding function $C(s, x_i)$?

$$x_{i+1} = 2x_i + s$$

x_{i+1}	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$s = 0$	0		1		2		3		4		5		6		7		8		9	
$s = 1$		0		1		2		3		4		5		6		7		8		9
x_{i+1}	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
$s = 0$	10		11		12		13		14		15		16		17		18		19	
$s = 1$		10		11		12		13		14		15		16		17		18		19
x_{i+1}	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59
$s = 0$	20		21		22		23		24		25		26		27		28		29	
$s = 1$		20		21		22		23		24		25		26		27		28		29

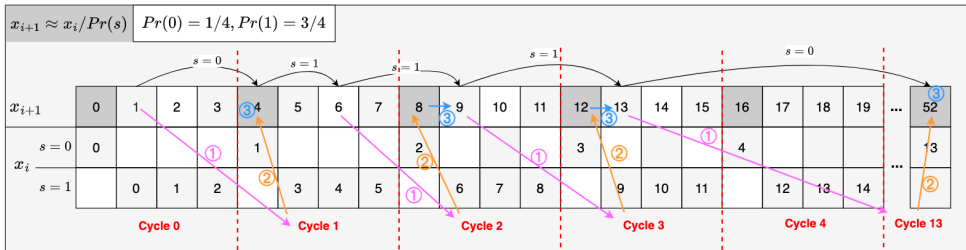
Fig.4: 101110

Exa.3: $Pr(0) = 1/4, Pr(1) = 3/4$

- How to encode 101110? What is the encoding function $C(s, x_i)$?
- How can we find the x -th appearance of symbol s ?
- $x_{i+1} = C(s, x_i) \approx x_i / p_s$

Some Notations

- $\mathcal{A} = \{s_1, s_2, \dots, s_k\}$ is the alphabet set
- s is the symbol, if it is used in subscript: s is the current symbol, $s - 1$ is the previous symbol, $s + 1$ is the next symbol
- F_s is the absolute frequency of symbol s
- $m = \sum_{s \in \mathcal{A}} F_s$ is the total frequency
- $c_s = \sum_{s \in \mathcal{A}} F_{s-1}$ is the cumulative frequency (c_{s+1} is the cumulative frequency of the next symbol)
- $p_s = F_s/m$ is the probability of symbol s
- x_i is the current state (next state: x_{i+1} , previous state: x_{i-1})



Given current state x_i and next symbol s . How to find next state x_{i+1} , which is the x_i -th appearance of symbol s ?

STEPS / EXAMPLE	$Pr(0) = 1/4, Pr(1) = 3/4$		General
	$s = 0$	$s = 1$	
Step ①: Find cycle number k	$k = \lfloor x_i / 1 \rfloor$	$k = \lfloor x_i / 3 \rfloor$	$k = \lfloor x_i / F_s \rfloor$
Step ②: Navigate to the beginning (first number) of that cycle	$4 \cdot k$	$4 \cdot k$	$m \cdot k$
Step ③: Find the position of the symbol s in that cycle	part 1: first of one + 0	part 1: + mod($x_i, 3$)	part 1: + mod(x_i, F_s)
- part 1: position of the symbol s in the sequence of the same symbol - part 2: how many other symbols before the symbol s	part 2: no other symbols + 0	part 2: only one other symbol (0) and with one of that symbol in one cycle + 1	part 2: only one other symbol (0) and with one of that symbol in one cycle + c_s
	$x_{i+1} = m \cdot k$	$x_{i+1} = 4 \cdot \lfloor x_i / 3 \rfloor + \text{mod}(x_i, 3) + 1$	$x_{i+1} = m \cdot \lfloor x_i / F_s \rfloor + \text{mod}(x_i, F_s) + c_s$

$$x_{i+1} = C(s, x_i) = m \cdot \lfloor x_i / F_s \rfloor + \text{mod}(x_i, F_s) + c_s$$

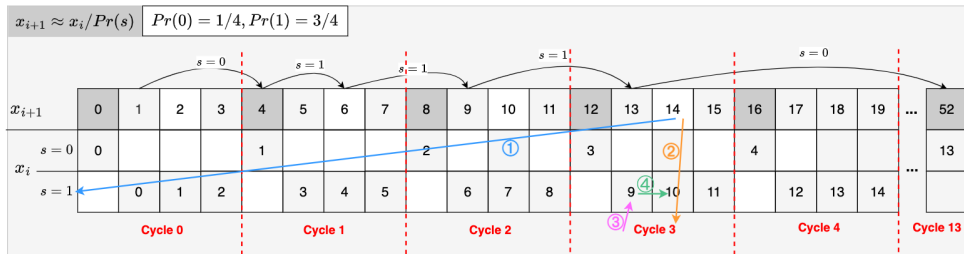
Fig.8

Encoding Function [2] [5]:

$$x_{i+1} = m \cdot \lfloor x_i / F_s \rfloor + \text{mod}(x_i, F_s) + c_s$$

implementation with python (dash app)

```
1 def rANS_encoder(symbol_counts, s_input):
2     cumul_freq = [] # c_s
3     sum_freq = 0 # m
4     x = 0 # initial state
5     output = [] # list to store encoded (symbols, state)
6
7     for count in symbol_counts:
8         cumul_freq.append(sum_freq)
9         sum_freq += count
10    for s in s_input:
11        F_s = symbol_counts[s]
12        c_s = cumul_freq[s]
13        x = sum_freq * (x // F_s) + (x % F_s) + c_s
14        output.append((s, x))
15
16    L_avg = math.ceil(math.log(x) / math.log(2.0)) / len(s_input)
17    return output, x, L_avg, entropy(symbol_counts)
```

Given current state x_i . How to find previous state x_{i-1} , where x_i is the x_{i-1} -th appearance of symbol s ?

STEPS / EXAMPLE	$Pr(0) = 1/4, Pr(1) = 3/4$		General
	$s = 0$	$s = 1$	
Step ① : Find the symbol s for the state x_i	if $x_i \bmod 4 = 0$	otherwise or if $x_i \bmod 4 = 1, 2, 3$	s such that $c_s \leq \text{mod}(x_i, m) < c_{s+1}$
Step ②: Find the cycle k of the current state x_i	$k = x_i / 4$	$k = \lfloor x_i / 4 \rfloor$	$k = \lfloor x_i / m \rfloor$
Step ③: Find the first number of that symbol s in that cycle k	$x_i / 4 \cdot 1$	$\lfloor x_i / 4 \rfloor \cdot 3$	$\lfloor x_i / m \rfloor \cdot F_s$
Step ④: Allocate to the exact position	$x_{i-1} = x_i / 4 \cdot 1 + 0 - 0$	$x_{i-1} = \lfloor x_i / 4 \rfloor \cdot 3 + 2 - 1$	$x_{i+1} = \lfloor x_i / m \rfloor \cdot F_s + \text{mod}(x_i, F_s) - c_s$

1. s such that $c_s \leq \text{mod}(x_1, m) < c_{s+1}$
2. $x_{i-1} = D(x_i) = \lfloor x_i / m \rfloor \cdot F_s + \text{mod}(x_i, F_s) - c_s$

Fig.9

Decoding Function [2] [5]:

— s

s such that $c_s \leq \text{mod}(x_i, m) < c_{s+1}$

— x_{i-1}

$$x_{i-1} = D(x_i) = \lfloor x_i / m \rfloor \cdot F_s + \text{mod}(x_i, F_s) - c_s$$

implementation with python (dash app)

```
1  def rANS_decoder(symbol_counts, x):
2      cumul_freq = [] # c_s
3      sum_freq = 0 # m
4      output = [] # List to store decoded symbols and states
5      # compute m and c_s
6      for count in symbol_counts:
7          cumul_freq.append(sum_freq)
8          sum_freq += count
9      while x > 0:
10         slot = x % sum_freq # Find the slot in the cumulative frequency range
11         s = bisect_right(cumul_freq, slot) - 1 # Binary search to find the symbol s (slot).
12         F_s = symbol_counts[s] # Frequency of the symbol
13         c_s = cumul_freq[s] # Cumulative count of the symbol
14         output.append((s,x)) # Append decoded symbol to the output
15         x = (x // sum_freq) * F_s + x % sum_freq - c_s # Update the state
16     return output, state
```

Streaming-rANS

Outline

- 2.2.1 Symbol spread function
- 2.2.2 Representation of information
- 2.2.3 Symmetric or Asymmetric

4. Streaming-rANS

- 4.1 Problem
- 4.2 Idea
- 4.3 Solution
 - 4.3.1 Renormalization
- 4.4 Implementation of Streaming-rANS
 - 4.4.1 Encoding
 - 4.4.2 Dncoding

Streaming-rANS

To avoid using large number arithmetic, in practice stream variants are used: which enforce by renormalization: sending the least significant bits of x to or from the bitstream (usually L and b are powers of 2). ^[4]

Example of the problem //TODO: add example

- ANS (Asymmetric Numeral Systems) uses a single integer x , known as the “state,” to represent the encoded data.
- Over time, as more symbols are encoded, the state x can grow arbitrarily large. This can lead to:
 - Overflow: If the value exceeds the precision limits of your system (e.g., a 32-bit or 64-bit register).
 - Inefficient Encoding: A larger x means you need more memory or bits to represent it.
- Using the C function from the previous section to encode succeeding symbols, we can encode a sequence into a very large natural number x , which can be stored as a length $\lfloor \lg(x) \rfloor + 1$ bit sequence. [2]

Renormalization in Range Coding

- Range Coding
 - 1
 - 2
- we use the similar idea for rANS
 - range $[L, H]$
 - Encoding: shrink the current state x until it is in the range $[L, H]$
 - Decoding: expand the current state x until it is in the range $[L, H]$
- In AC, renormalization is used to allow finite precision, an analogous approach should be used for ANS. Specifically, we will enforce x to remain in some fixed range I by transferring the least significant bits to the stream. For generality, let us transfer a base b digit, for example $b = 2$ means transferring single bits, while $b = 2^8$ means transferring complete bytes.

Methods from Jarek Duda: Method 1 - Decoding [2]

ANS decoding step from state x

$(s, x) = D(x)$ *# decode the current state x*

useSymbol(s)

while $x < L$ **do** *# if x is too small that not in the range $[L, H]$*

expand the x by reading the outstreamed digits back from the stream (an array)

$\text{expand}(x, b) = b \cdot x + \text{readDigit}() \rightarrow x$

end while

return the new state

What is L, H ?

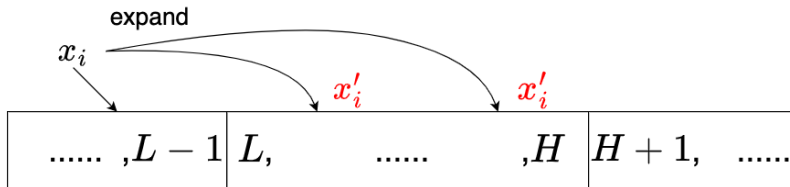


Fig 11

What is L, H ?

- $x_i = b \cdot x_{i-1} + \text{bit}$
- Either x_i in $[L, H]$ or x_{i-1} in $[L, H]$. Assume x_{i-1} in $[L, H]$.
 - $x_{i-1} \geq L$
 - $x_i \geq b \cdot L + \text{bit} \geq b \cdot L$
 - x_i can't be in $[L, H]$
 - derive: $H \leq b \cdot L - 1$
- $[L, H]$, where $H \leq b \cdot L - 1$ (Constraint 1)
- $I = \{L, \dots, b \cdot L - 1\}$, which is defined by Jarek Duda **b -unique interval** and $L \in \mathbb{N}$
- This will ensure that the loop in Method 1 will always return to the original value

Methods from Jarek Duda: Method 2 - Encoding ^[2]

shrink the current state x by stream out the least significant bit (x will be stored in binary and we take out the bit from right side)

ANS encoding step of symbol s from state x

once the current state x is greater than max value that we define later

while $x > \text{maxX}[s]$ **do**

write the least significant digit to the stream (stream could be an array in implementation)

writeDigit(mod(x , b))

shrink(x , b) = $\lfloor x/b \rfloor \rightarrow x$ *# shrink the x*

end while

$x = C(s, x)$ *# encode with the shrunk x*

What is $\max X[s]$?

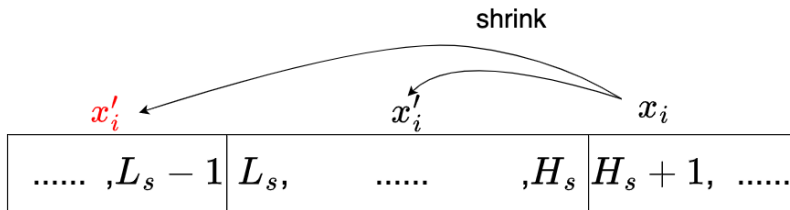


Fig 10

What is $\max X[s]$?

- to make sure not to jump over the range
- Assume that $x_i = H_s + 1$ and it will be shrinked to $x'_i = \lfloor x_i/b \rfloor$
- $x'_i = \lfloor x_i/b \rfloor = \lfloor (H_s + 1)/b \rfloor \geq L_s$
- $L_s \leq (H_s + 1)/b \leq \lfloor (H_s + 1)/b \rfloor < (H_s + 1)/b + 1$
- $H_s \geq b \cdot L_s - 1, \forall s \in \mathcal{A}$ (Constraint 2)

Final Form of $[L, H]$ in practice

- Constraints

$$[L, H] : \text{ where } H \leq b \cdot L - 1$$

$$[L_s, H_s] : H_s \geq b \cdot L_s - 1, \forall s \in \mathcal{A}$$

- In practice ($b = 2^1$): we choose $L = k \cdot m$, $L_s = k \cdot F_s$ ($k \in \mathbb{N}$)

$$\mathcal{I} = [km, 2 \cdot km - 1]$$

$$\mathcal{I}_s = [kF_s, 2 \cdot kF_s - 1] \text{ for } \forall s \in \mathcal{A}$$

implementation with python (dash app)

```
1  def rANS_stream_encoder(symbol_counts, s_input):
2      # Init: c_s, m, x, output, bit_stream (omit the code here)
3      for s in s_input:
4          F_s = symbol_counts[s]
5          c_s = cumul_freq[s]
6          L_s = F_s # k = 1
7          maxX_s = 2 * L_s - 1
8          out_bits = ""
9
10         while x > maxX_s:
11             out_bits += str(x % 2)
12             x = x // 2
13         x = (x // F_s) * sum_freq + (x % F_s) + c_s
14         bit_stream += out_bits
15         output.append((s, x, out_bits, bit_stream))
16
17     return output, x, bit_stream, L_avg, entropy(symbol_counts)
```

implementation with python (dash app)

```
1  def rANS_stream_decoder(symbol_counts, x, bit_stream, num_symbols):
2      # Init: c_s, m, x, output, bit_stream (omit the code here)
3      bit_stream = list(map(int, bit_stream.split(','))) # Convert bit_stream to a list of integers
4      for _ in range(num_symbols):
5          slot = x % sum_freq
6          s = bisect_right(cumul_freq, slot) - 1
7          F_s = symbol_counts[s]
8          c_s = cumul_freq[s]
9          output.append((s, x))
10         x = (x // sum_freq) * F_s + slot - c_s
11
12         while x < sum_freq and len(bit_stream) > 0:
13             x = x * 2 + bit_stream.pop()
14
15     return output, x
```

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

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